2026 Annual Teaching Plan
Natural Sciences and Technology: Grade 6

basic education
Department:
Basic Education
REPUBLIC OF SOUTH AFRICA

Life and living & Processing

Duration	CAPS Topics	Core Concepts, Skills and Values	Resources to Enhance Learning	Informal Assessment
2½ weeks	Photosynthesis	**Plants and food** • plants make their own food (glucose sugar) by a process called photosynthesis • photosynthesis takes place mainly in the leaves during photosynthesis the plant uses sunlight energy, carbon dioxide (from the air) and water to make glucose sugar • plants change some of the glucose sugar into starch which they store in their leaves, stems and roots, flowers, fruits and seeds **Plants and air** • during photosynthesis the plant uses carbon dioxide from the air and gives off oxygen into the air • animals, including people, use the oxygen from the air for breathing and give out carbon dioxide which is used by plants for photosynthesis	• Glucose powder, maize flour, iodine solution and plastic droppers • Examples of foods, such as cooked rice, flour, potato, bread, oil, boiled egg and cheese • Video clips from the internet	• Explain and illustrate how plants make food • Compare glucose sugar (such as glucose sweets) and starch (such as maize flour) according to their taste and colour • Test various foods for the presence of starch with iodine solution (e.g. cooked rice, flour, potato, bread, oil, boiled egg, cheese, etc.)
2 weeks	Nutrients in food	**Food Groups** • foods can be grouped according to their functions in the body and the main nutrients they supply: - foods for energy – carbohydrates - foods for growth and repair – proteins - foods for storing energy (in the form of body fats) and providing insulation and protection for nerves and organs - fats and oils - foods for building bones and teeth, and maintaining a healthy immune system - vitamins and minerals • most natural foods contain a mixture of more than one nutrient group • most processed (manufactured) foods have added salt, sugar, preservatives, flavourings and colourings	• Examples of different foods representing the different food groups • Food packaging with information about nutrients	• Classifying food into different food groups, namely carbohydrates, proteins, fats and oils and vitamins and minerals • State reasons why each food group is important in our diet • Read labels on food packaging to look for the nutrients and/or the additives in the food • Explain whether each of the additives make these products more or less healthy to eat • Carefully study various diets to evaluate if they contain all the food groups/whether they are a balanced diet • Explain why different portions of the different food groups are necessary for a balanced diet • Discuss various diseases caused by an unhealthy diet, such as tooth decay, obesity, diabetes or deficiency diseases
1½ week	Nutrition	**Balanced diets** • a diet refers to the selection of foods we eat everyday • a balanced diet contains sufficient quantities of food from all four nutrient groups, as well as water and fibre • some diseases can be related to diet	• Pictures and information about food-related illnesses	
3 weeks	Ecosystems and food webs	**Different ecosystems** • an ecosystem is an area where living and non-living things depend on each other in many different ways • there are many different ecosystems such as rivers, mountains, sea, rocky shore, ponds, wetlands, grasslands, forests and deserts, which support different kinds of living things **Living and non-living things in ecosystems** • in an ecosystem there are certain relationships between living things (plants, animals/people, microorganisms), and non-living things (air, water, sunlight, soil) in a particular area* (*Healthy ecosystems depend on sufficient biodiversity of plants, animals and their habitats*) **Food webs** • in an ecosystem plants and animals are connected by their feeding relationships. This is called a food web • a food web consists of: - plants (producers) which produce food for themselves and animals - animals (consumers) which are herbivores, carnivores, omnivores • microorganisms (decomposers) that break down dead plant and animal matter and return nutrients to the soil	• Pictures of ecosystems such as rivers, mountains, the sea, rocky shores, ponds, wetlands, grasslands, forests and deserts	• Describe different types of ecosystems on our planet • Identify an ecosystem, describe it and draw the feeding relationships (food webs) within it • Investigate an ecosystem in or near the school grounds. Mark out the area with sticks and string using the quadrant method, ensuring that you do not damage any of the plants or animals • Study both the living and non-living things within the ecosystem • Identify the possible threats to this ecosystem and possible ways to overcome them

1 week	Remediation, revision and consolidation
SBA (FORMAL ASSESSMENT)	• Practical task/investigation • Test

Matter and materials & Processing

Duration	CAPS Topics	Core Concepts, Skills and Values	Resources to Enhance Learning	Informal Assessment
1 week	Solids, Liquids and gases	**Arrangement of particles** • all matter (solids, liquids and gases) is made up of particles • the particles are arranged differently in solids, liquids and gases - in solids the particles are closely packed in a regular pattern – spaces between the particles are small and particles vibrate in one place - in liquids the particles are closely packed in no fixed pattern – spaces between the particles are small but particles can move around each other • in gases the particles are far apart from each other –spaces between the particles are big and particles move in all directions	• Video clips from the internet	• Draw and explain how particles are arranged in a solid, liquid and gas • Identify the three states of matter in everyday life • Describe solids, liquids and gases in terms of the arrangement of their particles
1 week	Mixtures	**Mixtures of materials** • a mixture consists of at least two different substances/materials mixed together • in some mixtures, the different substances are still clearly visible after mixing • the substances in such mixtures can be separated by physical means such as sieving, filtering, hand sorting, settling and decanting	• Video clips from the internet • Examples of materials and substances, such as salt, sand, sugar, tea leaves, peanuts, dried beans, coins, sweets, curry powder, grated cheese, milk, oil	• Explain and demonstrate the different ways in which solids, liquids and gases can be combined to form mixtures • Explain and demonstrate the different ways in which mixtures can be separated, such as sieving, filtering, hand sorting, settling and decanting • Investigate different solids to see whether they dissolve in water, including salt and sugar (soluble substances) as well as sand, mealie meal, flour, maize flour, samp, curry powder and custard powder (insoluble substances) • Investigate solutions to see whether we can recover the solute by filtering or settling followed by decanting, or evaporating the water (crystallisation) • Investigate and make sugar crystals • Explain different kinds of mixtures (including solutions) • Distinguish between soluble and insoluble substances • Recover the solute from the solvent and draw and write about the process
2 weeks	Solutions as special mixtures	**Solutions** • solutions are also mixtures • some solutions can be made by mixing a solid and a liquid together such as sugar and water, salt and water • solutions are uniform in appearance and the solid cannot be seen after mixing **Soluble substances** • soluble solids (solutes) can dissolve* in water (solvent) • the substances in solutions cannot be separated by sieving, filtering, hand sorting, settling and decanting • some solutes can be recovered (separated) by evaporating the solvent (such as recovering salt from sea water) • when substances dissolve, solute particles become dispersed in the spaces between the solvent particles **Saturated solutions** • a solution is saturated when no more solute can dissolve in a given amount of solvent **Insoluble substances** • Some solids will not form a solution in water (insoluble solids)	• Video clips from the internet • Examples of materials and substances such as salt, sugar, sand, mealie meal, flour, maize flour, samp, curry powder, custard powder • Measuring cylinders, funnels, filter paper, beakers, evaporating dish, salt, food colouring	
1 week	Dissolving	**Rates of dissolving** • Factors that affect the rate (time taken) of dissolving - temperature of the mixture - stirring or shaking the mixture • grain size of the solute	• Glass beakers/small yoghurt tubs/clear containers, a stopwatch or clock with a second hand, thermometers, measuring spoons, ice cream sticks, salt (coarse and fine) • Video clips from the internet	• Investigate the difference between melting and dissolving • Investigate, measure and draw graphs of the time taken to dissolve a solute in hot or cold water when stirring/shaking or not, and using coarse or fine salt

				• Tell what factors affect the rate of dissolving
2 weeks	**Mixtures and water resources**	**Water pollution** • in the environment, many things mix or dissolve in water • water can be polluted by - insoluble substances, such as oil, plastics, tyres, tins, glass, toilet waste - soluble substances such as soaps, fertilizers, insecticides, acids and other poisons - living germs from toilet waste causing water-borne illnesses such as diarrhoea **Importance of wetlands** • natural wetlands are important for removing soluble and insoluble substances from water • acting like sponges and regulating the flow of water	• Texts for reading about water pollution • Video clips from the internet • Pictures of different kinds of polluted water • Sources about, and pictures of, different wetlands	• Discuss pollution and where it comes from. • Identify three main categories of pollutants found in water and explain how you think they entered/ended up in the water. • Explain why wetlands are so important • Research the different wetlands in South Africa
2 weeks	**Processes to purify water**	**Clean water** • a clean supply of water is important for people, plants and animals water can be cleaned by processes such as sieving, filtering, settling, decanting, boiling and adding chemicals to kill germs • municipal water is cleaned before and after we use it	• Sieves, filter paper, funnels, containers, a kettle and water purification tablets (if possible)	• Design, make and evaluate a system to process and purify dirty water • Investigate how to best purify dirty water in class and/or at home
1 week	**Remediation, revision and consolidation**			
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Energy and change & Systems and control

Duration	CAPS Topics	Core Concepts, Skills and Values	Resources to Enhance Learning	Informal Assessment
3 weeks	**Electric circuits**	**A simple circuit** • an electric circuit is a system for transferring energy • a simple circuit always has the following components: - source of energy (such as a cell/battery) - conducting material (such as wires) - device (such as a light bulb, buzzers, motors) for changing electricity into a useful output energy • a circuit is a complete, unbroken pathway for electricity • a switch can be added to break or complete the circuit pathway **Circuit diagrams** • symbols are used when drawing circuit diagrams	• Equipment such as cells/batteries, conducting wires, light bulbs and switches • Video clips PhET simulations from the internet	• Investigate different ways of making a simple circuit • Investigate, design and make a switch to control the circuit • Investigate and understand how the different components function and what symbols are used to represent them in a circuit diagram. • Draw simple closed electrical circuit diagrams using the symbols for the various components
2 weeks	**Electrical conductors and insulators**	**Conductors** • some materials conduct electricity and are called conductors - most metals, especially copper, conduct electricity **Insulators** • some materials do not conduct electricity and are called insulators • most non-metals, such as plastics, do not conduct electricity	• Cells/batteries, conducting wires, light bulbs and switches • Different materials including metal paper clips, nails, wire, steel wool, coins, plastic, glass, ceramic, cardboard, paper, wood, rubber, chalk, plastic insulated wires, rubber gloves used by electricians • Video clips and PhET simulations from the internet	• Investigate what conductors and insulators are by testing different materials (such as metal paper clips, nails, wire, steel wool, coins, plastic, glass, ceramic, cardboard, paper, wood, rubber and chalk) in an electric circuit to see if they are conductors or insulators and recording the results in a table • Identifying where electrical insulators are used, e.g.: insulating wires, rubber gloves, glass and ceramic insulators on power lines

1 week	**Systems to solve problems**	**Using electric circuits** • electric circuits are often used to solve problems that require energy, such as street lighting, alarms, electric gates, traffic lights, fans and heaters • electric circuits can also be used in models and toys	• Basic components for a circuit, including components such as cells, light bulbs, conducting wires, buzzers and switches. • Video clips and PhET simulations from the internet	• Design, make, evaluate and present a system that uses a circuit to produce movement, light, sound or heat in a structure such as: • A steady hand game, house, light house or a toy • The circuit should include components such as cells, light bulbs, buzzers and switches.
2 weeks	**Mains electricity**	**Fossil fuels and electricity** • fossil fuels were formed in the Earth's crust millions of years ago from dead plants and animals • coal, oil and natural gas are fossil fuels • in South Africa coal is mostly used as a fuel in power stations • coal was formed from fossilised plants which got their energy from the Sun originally • in a power station coal is used to boil water, the steam turns a turbine which turns a generator, which produces electricity • fossil fuels are non-renewable resources **Cost of electricity** • electricity is costly because - it requires infrastructure including coal mines, transport, power stations, pylons, substations, wiring - some electrical appliances require more electricity than others (heating appliances use the most) • the more electricity we use the more we pay and the more coal is used up • we can save energy in many ways including using energy saving light bulbs and solar water heaters **Renewable ways to generate electricity** • people are looking for renewable ways to generate electricity	• Pictures and video clips of fuels and their various uses • Pictures to show how electricity is generated in a coal- fired power station • Examples of electrical appliances • Pictures and information of renewable ways to generate electricity, including examples of wind power generators, solar power generators and hydro-electric power generators	• Explain the steps that outline the process to make electricity from coal • Use diagrams to trace and explain the flow of electrical energy in a sequence from an appliance, such as from your TV set to the coal-fired power station and back to the original source, the Sun. • Examine labels (in adverts or on real electrical appliances) to find out how much power they require in a certain time (e.g. a kettle, a radio, a TV, an iron, a hot plate, charging a cell phone, etc. (most kettles require more than 2 000 W, while a radio requires just more than 15 W) and make comparisons • Explain different ways to save electricity, from small actions to larger actions • Research and write about renewable ways to generate electricity, including in wind power generators, solar panels (photovoltaics), hydro-electric power generators, biomass and geothermal energy.
1 week	**Remediation, revision and consolidation**			
SBA (FORMAL ASSESSMENT)	• Practical task/investigation • Test			

Planet Earth and beyond & Systems and Control

Duration	CAPS Topics	Core Concepts, Skills and Values	Resources to Enhance Learning	Informal Assessment
2½ weeks	**The Solar System**	**The Sun, Planets and Asteroids** • the Sun (a star) is at the centre of our Solar System • there are eight planets and the asteroid belt (Mercury, Venus, Earth, Mars, Asteroid Belt, Jupiter, Saturn, Uranus, and Neptune) in orbit around the Sun • each planet has its own - features, size, orbit and position in relation to the Sun, composition (rocky and gas planets) and number of moons (some have no moons) • the planets and Asteroids take different amounts of time to revolve around the Sun* (*It is not necessary to memorize exact numbers of size of planets, number of moons, and distance from the Sun*) **Moons**	• Detailed pictures and models of the Solar System (not to scale) • Pictures of the Moon • Video clips from the internet	• Research/read information about the planets, focusing on size, distance from the Sun, average temperature, number of moons, atmosphere, and any other features • Making models of the Solar System (not to scale) considering the position in relation to the Sun and the size and features of the planets • Describe and draw the objects in our Solar System

		• moons, including our Moon do not give out their own heat and light • our Moon can be seen from Earth because the light from the Sun shines onto its surface • on the Moon we can see craters, lighter areas which are mountains, and darker areas which are flat plains		
1½ weeks	**Movements of the Earth and planets**	**Rotation (Earth)** • in our Solar System, each planet rotates (spins) on its own axis - the planet Earth is spinning, and one complete rotation takes about 24 hours. We experience this as a day and a night - during rotation the side of the Earth facing the Sun experiences daytime, and the opposite side of the Earth experiences night-time **Revolution (Earth)** • all planets also revolve (travel) around the Sun in their own orbits • planet Earth revolves around the Sun in its own orbit (pathway), and one complete revolution takes 365 ¼ days. We experience this as a year	• Models and a light source such as torch, lamp or candle to demonstrate the movements of the Earth	• Demonstrate the movements (rotation and revolution) of the Earth using models and body movements • Demonstrates how daytime and night-time occur using a model of the Earth and a light source • Draw and write about the rotation of the Earth in relation to the Sun: How daytime and night- time occur
1½ weeks	**The movement of the Moon**	**Rotation (Moon)** • the Moon is spinning and one complete rotation takes about 28 days **Revolution (Moon)** • the Moon revolves around the Earth and one revolution also takes about a month (about 28 days) • together, the Earth and the Moon revolve around the Sun	• Models and a light source such as torch, lamp or candle to demonstrate the movements of the Moon	• Demonstrate the rotation and revolution of the Moon around the Earth using models and body movements • Draw and write about the rotation of the Moon in relation to the Earth and the solar system • Drawing a table of comparison between the Sun, the Earth and the Moon including shape, composition, size, movement in relation to other space objects and the ability to produce light
1 week	**Systems for looking into space**	**Telescopes** • telescopes are used to look into space and gather information • South Africa has built and uses some of the largest telescopes	• Pictures and information about telescopes	• Identify the different telescopes and how they work • Doing a case study about telescopes such as simple telescopes, SALT and SKA
1½ weeks	**Systems to explore the Moon and Mars**	**Vehicles used on the Moon** • a few people have visited the surface of the Moon and explored it using a vehicle called a Moon Rover **Vehicles used on Mars** • robots called Mars Rovers have been used to visit and explore the surface of Mars (people have not yet visited Mars)	• Pictures of the Moon and Mars Rovers • Video clips from the internet • Apparatus for building models of rovers that can move on different surfaces • Measuring tapes or metre sticks	• Describe the vehicles used to explore the Moon and Mars • Researching the key features and purposes of the Moon and Mars rovers including the components, build and systems used for energy and communications • Design, make and evaluate a model of one of the vehicles moving with wheels and axles
1 week	**Remediation, revision and consolidation**			
SBA (FORMAL ASSESSMENT)	• Test			

Major Process and Design Skills

The teaching and learning of Natural Sciences and Technology involves the development of a range of process and design skills that may be used in everyday life in the community and in the workplace. Learners also develop the ability to think objectively and use a variety of forms of reasoning while they use these skills. Learners can gain these skills in an environment that taps into their curiosity about the world, and that supports creativity, responsibility and growing confidence.

The following are the cognitive and practical process and design skills that learners will be able to develop in Natural Sciences and Technology.

1. ***Accessing and recalling information*** – being able to use a variety of sources to acquire information, and to remember relevant facts and key ideas, and to build a conceptual framework
2. ***Observing*** – noting objects, organisms and events in detail
3. ***Comparing*** – noting similarities and differences between things
4. ***Measuring*** – using measuring instruments such as rulers, thermometers, clocks and syringes (for volume)
5. ***Sorting and classifying*** – applying criteria to sort items into a table, mind-map, key, list or other format
6. ***Identifying problems and issues*** – being able to articulate the needs and wants of people in society
7. ***Raising questions*** – being able to think of and articulate relevant questions about problems, issues and natural phenomena
8. ***Predicting*** – stating, before an investigation, what the learner thinks the results will be for that investigation
9. ***Hypothesising*** – putting forward a suggestion or possible explanation to account for certain facts. A hypothesis is used as a basis for further investigation that will prove or disprove the hypothesis
10. ***Planning investigations*** – thinking through the method for an activity or investigation in advance. Identifying the need to make an investigation a fair test by keeping some things (variables) the same, while other things will vary
11. ***Doing investigations*** – this involves carrying out methods using appropriate apparatus and equipment, and collecting data by observing and comparing, measuring and estimating, sequencing, or sorting and classifying. Sometimes an investigation has to be repeated to verify the results
12. ***Recording information*** – recording data from an investigation in a systematic way, which includes drawings, descriptions, tables and graphs
13. ***Interpreting information*** – explaining what the results of an activity or investigation mean (this includes reading skills)
14. ***Designing*** – showing (e.g. by drawing) how something is to be made, taking into account the design brief, specifications and constraints
15. ***Making/constructing*** – building or assembling an object using appropriate materials and tools and using skills such as measuring, cutting, folding, rolling and gluing
16. ***Evaluating and improving products*** – using criteria to assess a constructed object and then stating or carrying out ways to refine that object
17. ***Communicating*** – using written, oral, visual, graphic and other forms of communication to make information available to other people.